

Title of Invention:

Vigilant AI-Powered Fall Detection with Embedded LiDAR Technology

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Type of Applicant:

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SMALL ENTITY

☐

CORPORATE

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Abstract:

- Our paper introduces a pioneering method for fall detection, seamlessly integrating LiDAR technology with selectively activated cameras to ensure accurate detection while maintaining user privacy and comfort.
- Leveraging the RPLIDAR A1 LIDAR sensor and sophisticated AI algorithms on Jetson Nano, real-time fall detection is achieved without necessitating users to wear any devices. The system components, including NVIDIA JetPack, RPLIDAR SDK, OpenPose, YOLO, OpenCV, and Python Scripting with SMTP Libraries etc, work harmoniously together to achieve efficient and reliable fall detection
- The system's outputs include the generation and transmission of alert messages containing pertinent information and captured images, securely sent through our app.
- This innovative solution boasts several advantages, including its wearable-free design, selective camera activation preserving privacy, and enhanced user trust due to its non-intrusive nature.
- Moreover, the system proves beneficial for bedridden individuals, providing added security and care.
- Overall, this invention represents a significant advancement in fall detection technology, offering a user-centric solution that prioritizes comfort, privacy, and effectiveness

Specifications:

- The LIDAR Fall Detection System with Selective Camera Activation introduces an innovative method for fall detection, seamlessly integrating LIDAR technology with selectively activated cameras to ensure accurate detection while maintaining user privacy and comfort.
- By leveraging the RPLIDAR A1 LIDAR sensor, sophisticated AI algorithms running on the Jetson Nano platform enable real-time fall detection without necessitating users to wear any devices.
- The designed system components, including NVIDIA JetPack, RPLIDAR SDK, OpenPose, YOLO, OpenCV, and Python Scripting with SMTP Libraries, work harmoniously together to achieve efficient and reliable fall detection.

- Of notable mention are the privacy measures implemented, such as on-device image processing and secure communication protocols, which effectively safeguard user data.
- The system's outputs include the generation and transmission of alert messages containing pertinent information and captured images, securely sent through our app.
- This innovative solution boasts several advantages, including selective camera activation preserving privacy, and enhanced user trust due to its non-intrusive nature.
- Overall, this invention represents a significant advancement in fall detection technology, offering a user-centric solution that prioritizes comfort, privacy, and effectiveness.

1.Describe the Novel Feature:

- Unlike traditional fall detection systems that rely on wearable devices, our system utilizes Lidar technology for fall detection.
- This eliminates the need for users to wear any devices, enhancing comfort and compliance. The system incorporates a camera that is activated only when a fall is detected.
- This selective activation ensures privacy by minimizing the feeling of being constantly monitored while still providing critical visual information to administrators when needed.
- By eliminating the need for wearable devices and minimizing continuous camera surveillance, our system enhances user trust and comfort.
- Users are less likely to feel intrusive monitoring, leading to improved user acceptance and adoption of the fall detection system.
- Upon detection of a fall, the system immediately sends the captured image to administrators or nurses through our app.
- As our app sends the image with time it enables swift action and intervention, potentially saving the individual from further harm or facilitating timely medical assistance.
- These features represent significant advancements in fall detection technology, addressing user concerns about comfort, privacy, and trust while providing administrators with valuable visual data in the event of a fall.

2. Drawbacks of the existing state-of-art and how the drawbacks have been overcome and advantages of your invention

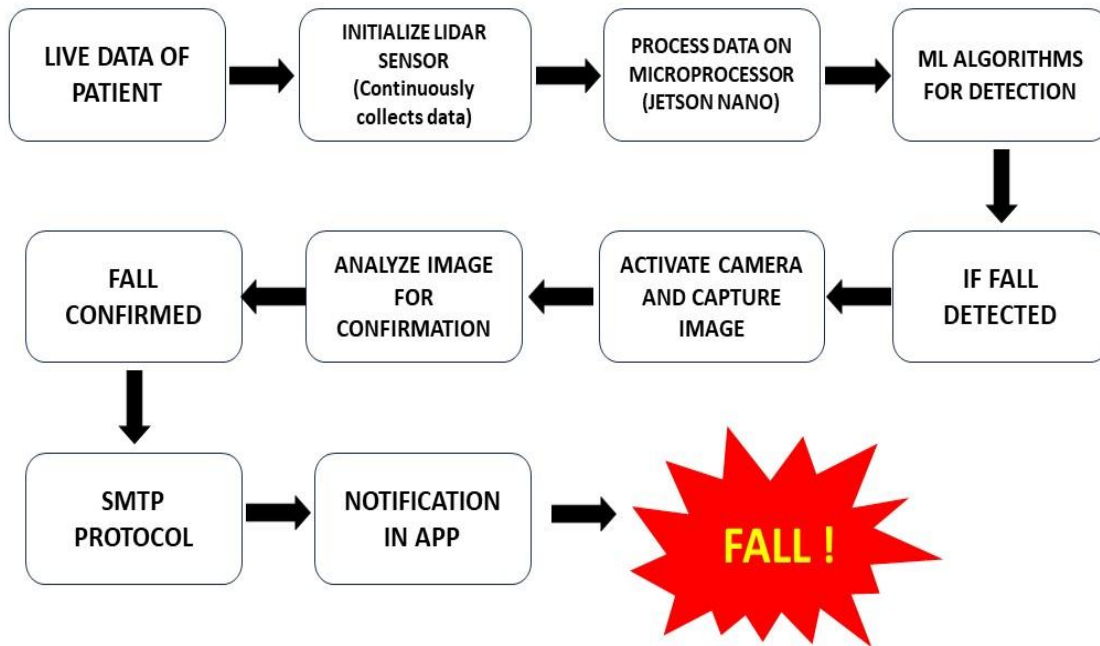
Disadvantages of current advanced technology:

- Current fall detection systems often have inaccurate detection capabilities, leading to false alarms or slips due to their reliance on sensor technology less accurate.
- Many existing systems require individuals to wear specific sensors, leading to compliance issues and discomfort, especially among old age users who may object to the use of any additional accessories.
- Moreover, individuals may remove wearable devices, such as watches, during activities such as bathing, compromising continuous monitoring and reducing effectiveness.
- Continuous surveillance via cameras or wearables raises privacy concerns, potentially affecting user trust and acceptance due to perceived intrusion into personal space.
- In addition, continuous monitoring may cause irritability or mental disorders in people being monitored.

Benefit from our innovation:

- Leveraging Lidar technology ensures accurate fall detection based on changes in distance and altitude, reduces false alarms and improves reliability.
- Because our system works without the need for a wearable device, we can effectively monitor patients without the risk of them taking it off, thereby improving acceptance and compliance by users by eliminating the discomfort and stigma associated with wearing additional accessories.
- Activating the camera only when a fall is detected reduces privacy concerns, ensuring that sensitive information is only recorded and transmitted when necessary.
- This approach reduces mental discomfort caused by constant monitoring, improving user comfort and technology acceptance.

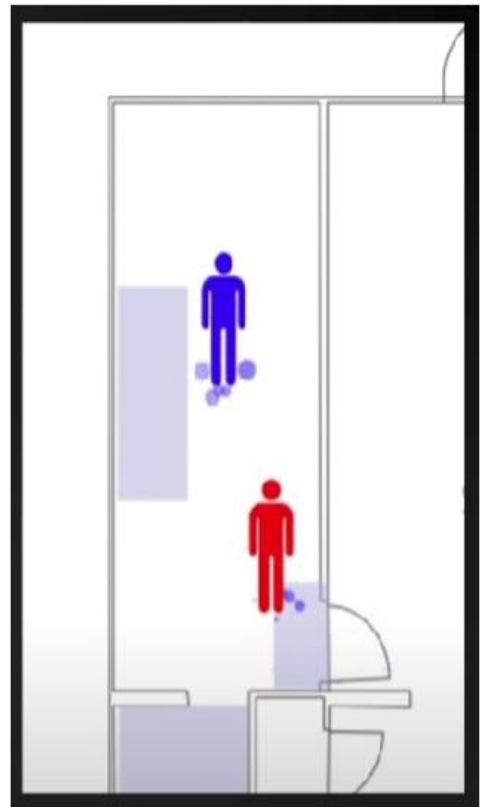
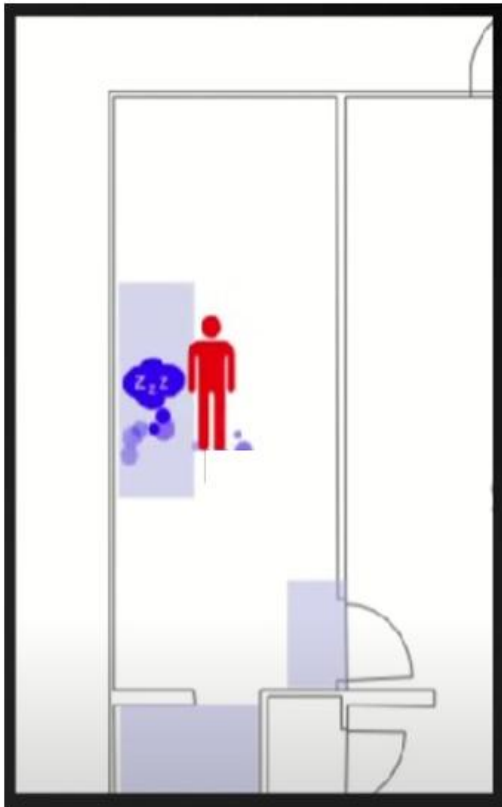
3. Brief Description of Figures



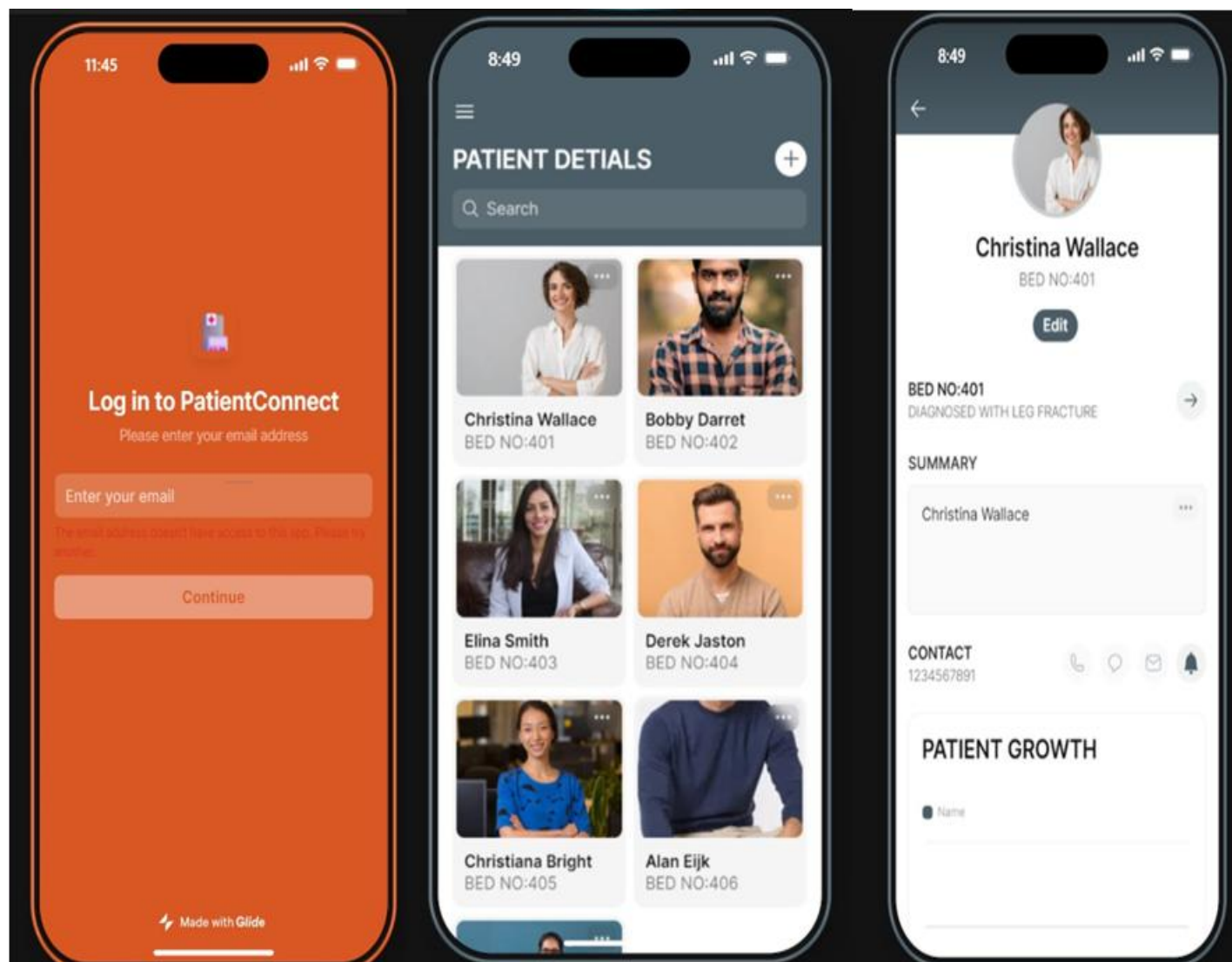
PatientConnect

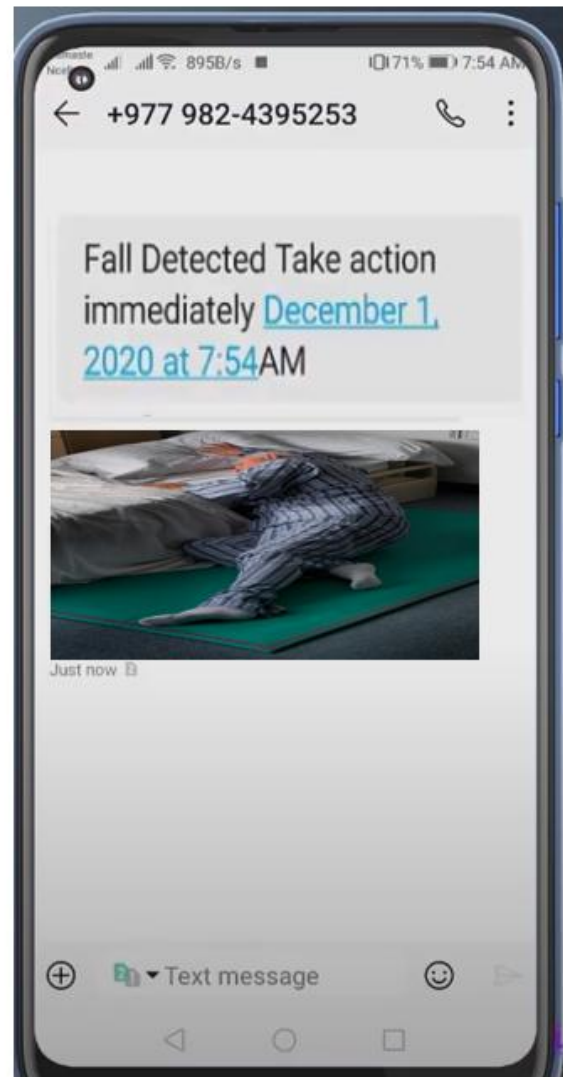
Description:

- ✓ PatientConnect is an advanced mobile application which we have designed to provide healthcare professionals and caregivers with advanced tools for real-time patient monitoring and seamless communication.
- ✓ With a user-friendly interface and powerful features, PatientConnect ensures effective health monitoring and facilitates streamlined interactions.



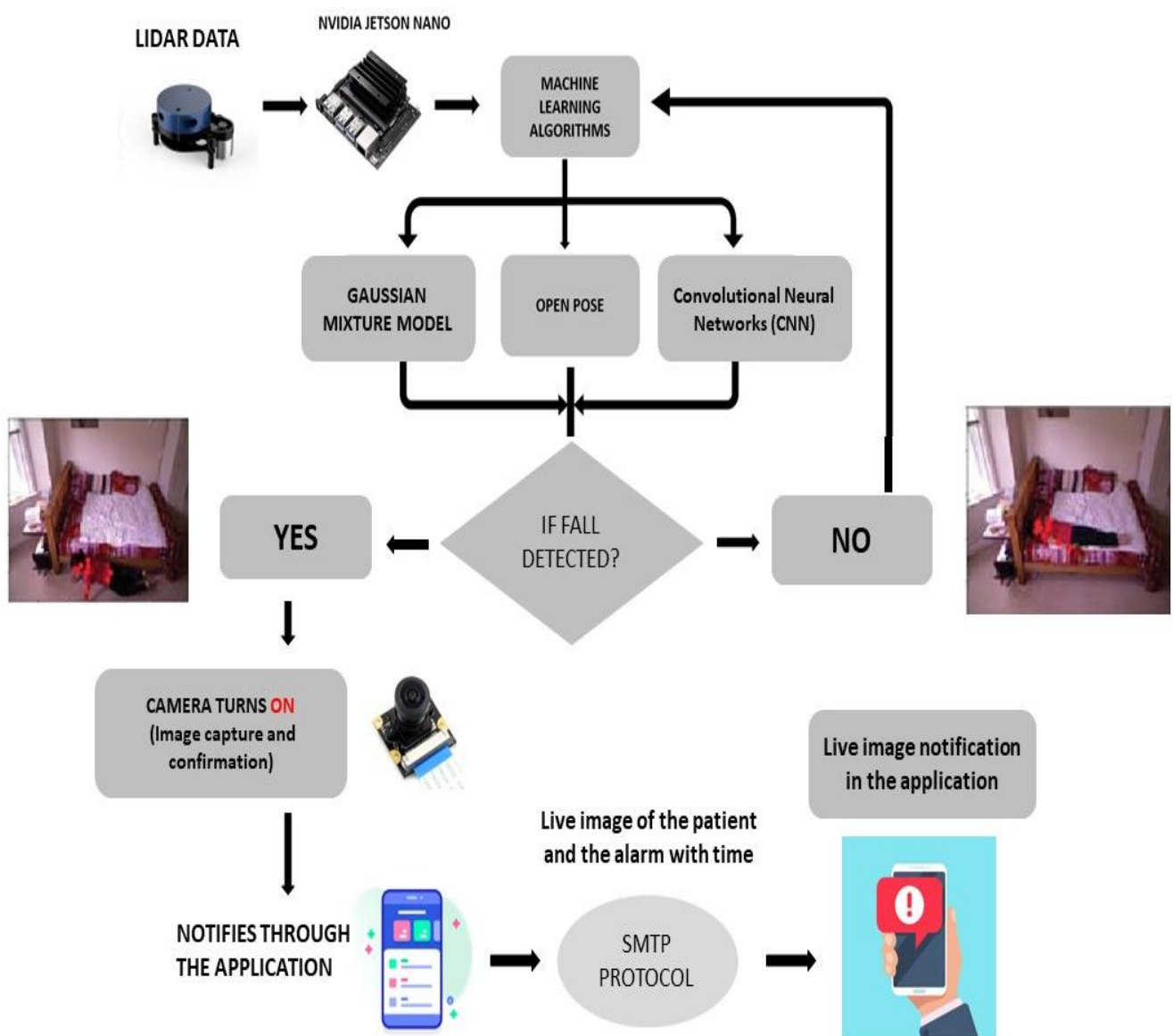
The RPLIDAR A1 detects falls, complemented by CNN-based object detection distinguishing patients by the colour of their hospital gown to avoid false detections of nurses or caretakers. Upon fall detection, the system sends an alert message containing the captured image through the designated app. Here, the blue-coloured person represents the patient, and the red-coloured person represents the caretaker.





Detailed Description

1.THE INVENTION OUTLINE [Block Diagram of all components or Process Diagram]



Input:

RPLIDAR A1 data provides depth information as well as accurate distance measurements. Pictures from Camera: When a fall is detected, pictures are captured for additional study.

Components:

1. **Jetson Nano** is an edge computing platform that uses GPU acceleration to run AI algorithms quickly.
2. **RPLIDAR A1** is a 2-D LIDAR sensor that can detect depth and estimate distance accurately.
3. **NVIDIA JetPack**: An operating system that includes crucial libraries and frameworks for AI development on Jetson Nano.
4. **RPLIDAR SDK**: Allows communication with the RPLIDAR A1 sensor, which extracts depth data for the fall detection algorithm.
5. **OpenPose**: A pre-trained model for real-time fall detection using human pose estimate.
6. **YOLO (You Only Look Once)**: A lightweight object identification system for identifying likely things associated with falls.
7. **OpenCV**: A library for taking pictures and processing.
8. **Gaussian Mixture Model (GMM)**: To distinguish normal activities from falls by analysing deviations from learned patterns, enhancing accuracy and reliability.
9. **Python Scripting with SMTP Libraries**: Creates alert messages that include relevant details and the captured image, then sends them via email or messaging API.

Process:

1. **Jetson Nano Setup**: The NVIDIA JetPack operating system is used to run Jetson Nano.
2. **Communication with RPLIDAR A1**: The RPLIDAR SDK allows RPLIDAR A1 to communicate with Jetson Nano and provide depth information.
3. **Fall Detection with OpenPose**: OpenPose does real-time fall detection using human pose estimation.
4. **Object Detection with YOLO**: YOLO detects possible fall-related objects.

5. **Image Capture with OpenCV:** When a fall is detected, OpenCV takes pictures using the associated camera.

6. Python scripting generates **alert messages** that include the **captured image**.

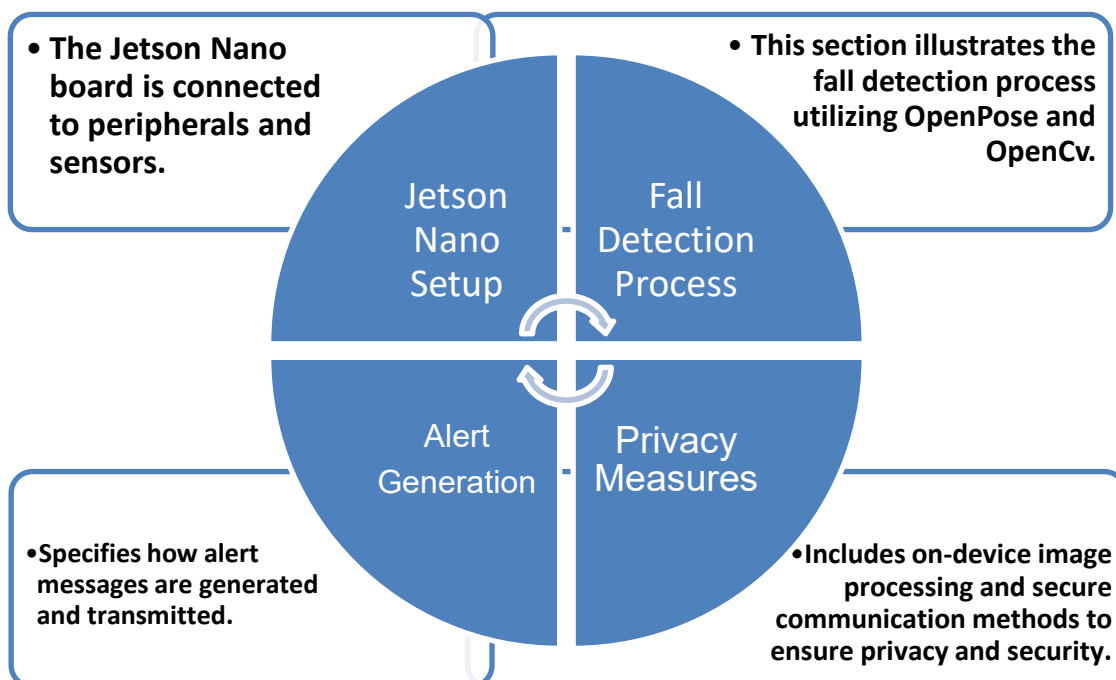
7. **Privacy Measures Implementation:** On-device image processing eliminates the need for external servers, while secure communication protocols ensure that alerts are securely transmitted.

Output:

Alert Messages: Created alert messages with pertinent information and the obtained image.

Secure Transmission: Alerts are securely communicated via email or messaging API.

Figure Explanation:



By combining these components and processes, the invention delivers accurate fall detection, lowers false alarms, ensures privacy, and improves user acceptability and compliance.

2.SPECIFIC COMPONENTS EXPLAINED

RPLIDAR A1:

RPLIDAR A1, plays an important role in the fall detection system by providing accurate distance measurements in a 360-degree field of view. When integrated with additional sensors, such as triaxial accelerometer for motion detection and camera for visual verification, RPLIDAR A1 improves the accuracy and reliability of fall detection algorithms. This integration ensures complete coverage and durability against various environmental conditions, including low or obscured light. Additionally, the versatility of RPLIDAR A1 allows for seamless deployment in both indoor and outdoor environments. Its high-resolution scanning capabilities enable detailed analysis of the environment, further enhancing the system's effectiveness in detecting falls. Additionally, the compact form factor and low power consumption of RPLIDAR A1 make it suitable for embedded applications, such as wearable fall detection devices. Overall, RPLIDAR A1 serves as the primary sensor in a fall detection system, helping to improve the safety and health of people at risk of falling.



NVIDIA Jetson Nano:

Jetson Nano forms an essential element in the field of fall detection systems, contributing significantly to their performance and efficiency. With high-performance computing, this compact device facilitates real-time processing of sensor data critical for fall detection. Equipped with a powerful GPU, Jetson Nano accelerates the execution of deep learning models such as convolutional neural networks (CNN), allowing rapid analysis of image data from sources such as Open Pose.

By leveraging edge computing, it allows the 's fall detection algorithms to run directly on the device, minimizing latency and ensuring rapid response in emergencies. Additionally, the Jetson Nano's low power consumption makes it well-suited for continuous monitoring applications. Its seamless integration with various sensors, including LiDAR and cameras, enriches input data for more accurate detection algorithms.



3.ALTERNATE EMBODIMENTS

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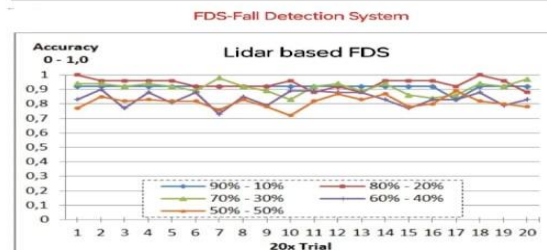
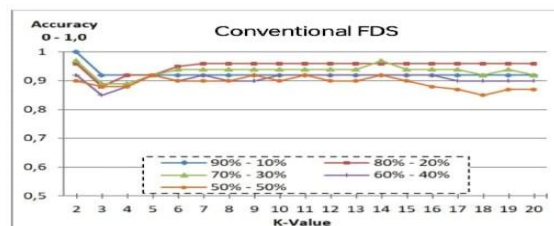
4.EXPERIMENTAL RESUTS & GRAPHS

Sample pictures from data set



Efficiency of Lidar based FDS

Learning Approach	Overall Accuracy	Precision (Falling)	Precision (Lying)	Precision (Standing Up from Lying)	Precision (On All Fours)	Precision (Other Activity)
Random forest	81.97%	66.67%	80.17%	90.00%	100.00%	83.12%
Artificial neural network	86.33%	14.29%	87.24%	71.12%	75.29%	87.99%
Decision tree	81.87%	50.00%	80.51%	50.66%	76.92%	84.60%
Multway decision tree	83.12%	0.00%	83.36%	63.34%	49.32%	85.90%
Support vector machine	76.50%	0.00%	67.34%	0.00%	0.00%	84.45%
k-NN	98.32%	87.72%	98.42%	96.71%	96.09%	99.09%
Gradient boosted trees	84.95%	0.00%	84.52%	67.20%	51.24%	88.61%
ID3	68.89%	0.00%	66.47%	66.47%	59.55%	70.39%
Decision stump	50.96%	0.00%	44.36%	0.00%	0.00%	51.20%
CHAID	68.87%	0.00%	66.44%	60.00%	59.55%	70.39%
AutoMLP	86.30%	30.43%	85.23%	70.72%	52.75%	91.45%
Linear regression	77.80%	0.00%	71.56%	0.00%	0.00%	82.58%
Vector linear regression	50.66%	0.00%	0.00%	0.00%	0.00%	50.66%
Random tree	77.29%	0.00%	76.04%	0.00%	0.00%	78.10%
Naïve Bayes	27.23%	5.69%	69.85%	12.95%	5.53%	0.00%
Naïve Bayes (kernel)	80.28%	4.00%	80.90%	47.87%	23.97%	85.08%
Linear discriminant analysis	50.66%	0.00%	0.00%	0.00%	0.00%	50.66%
Quadratic discriminant analysis	50.66%	0.00%	0.00%	0.00%	0.00%	50.66%
Deep learning	82.13%	33.33%	79.23%	56.75%	61.11%	86.61%



5.APPLICATION AREAS

- **Elderly care facilities:** Implement the system in a nursing home, assisted living facility, to provide pre-fall detection and timely assistance to seniors. This can help reduce your risk of injury and improve your overall safety and well-being.
- **Hospitals and Rehabilitation Centers:** Using the system in hospital rooms, rehabilitation centers or recovery rooms to monitor patient movements and detect falls, especially in patients with mobility issues or receiving physical therapy.
- **Home Healthcare:** The system is provided as a home monitoring solution for seniors living alone or suffering from chronic diseases. Family members or caregivers can be alerted in the event of a fall and remotely access camera footage, providing peace of mind and immediate assistance.
- **Emergency response systems:** Integrate fall detection systems with first responders to provide rapid intervention and medical care during fall emergencies, especially for people at high risk or with certain medical conditions.
- **Workplace Safety:** Deploy our systems in industrial plants, construction sites or warehouses to improve workplace safety by detecting and eliminating fall hazards in real time. This can help prevent workplace accidents and improve overall health and safety practices.
- **Sports and Fitness Monitoring:** Apply our systems to sports training facilities, gyms or outdoor recreation areas to track athletes' movements and detect potential injuries or falls while training or competing.

6.PUBLIC DISCLOSURE:

--NIL--